



Modeling drivers' acceptance of augmented reality head-up display in connected environment[☆]

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ABSTRACT

Investigating the acceptance of augmented reality head-up displays (AR-HUD) in connected environments is an important research issue as there is great potential to enhance driving safety for both manual and autonomous driving. This study investigated drivers' acceptance of AR-HUD with the technology acceptance model (TAM) with extended variables, including trust, information service quality, display service quality, perceived distraction, and driving skills. The answers to a survey completed by 388 drivers were analyzed with structural equation modeling (SEM) to verify the reliability and validity of the measurement model and explore the relationships between the observed variables. The results show that all the hypotheses of foundational TAM are supported. Further analysis has indicated that: (1) drivers' trust has a positive effect on perceived usefulness, attitude, and intention to use; (2) display service quality affects perceived ease of use positively, while information service quality has a positive influence on trust and intention to use; (3) perceived distraction has a significant negative impact on perceived ease of use and trust, while the negative effect on perceived usefulness is not significant; and (4) drivers' subjective driving skills have a positive influence on perceived ease of use and a negative impact on perceived usefulness. Further analysis attempts to explain the causes of the influence relationship between variables, and suggestions are proposed to improve the drivers' perceived usefulness and perceived ease of use of AR-HUD. The study provides valuable insights into the factors that significantly affect drivers' intention to use AR-HUD systems in theory and provides suggestions to AR-HUD designers on system design.

1. Introduction

As a new form of in-vehicle human-vehicle interaction, the head-up display (HUD) projects visual information into drivers' forward-looking visual field and reduces the eye-off road frequency and duration [1]. Previous studies have empirically demonstrated the positive effects of HUDs compared to head-down displays (HDDs) in terms of the performance of primary and secondary driving tasks [2,3]. During manual driving, HUD can effectively reduce drivers' eyes-off-the-road time and re-accommodation demands to avoid visual distraction [4] and reduce traffic accidents related to drivers' visual errors [5,6]. In autonomous driving, the information about the driving status and surroundings provided by HUD enhances drivers' situation awareness and the efficiency of communicating with the autonomous vehicle, improving drivers' response and take-over ability [7,8]. Currently, HUDs with

active safety warnings have been classified as a component of advanced driver assistance systems (ADAS) and have become increasingly popular. The HUD system is considered to pave the way for the development of connected and autonomous driving [9]. It is estimated that the annual growth rate of HUDs will be as high as 29.91 %, and almost one-third of vehicles will be equipped with a HUD system by 2024 [10].

In recent decades, the sensing and wireless technologies are growing rapidly, and have facilitated the application of a group of technologies of vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I). V2V communication enables vehicles to wirelessly exchange information about their speed, location, and heading, and allows vehicles to broadcast and receive omni-directional messages [11]; V2I communication provides information such as advisories from the infrastructure to the vehicle informing the driver of safety, mobility, or environment-related conditions [12]. Hence, the emerging driving environment, i.e., the

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connected environment, that allows effective data exchange among elements and actors of the transport system may be more prevalent in the future. The information on surrounding traffic and areas beyond the visual range can be provided to improve drivers' knowledge of the driving situation and avoid dangerous incidents in the connected environment [13]. Meanwhile, augmented reality head-up display (AR-HUD) is emerging as the latest generation of in-vehicle HUD display technology [14]. AR refers to the combination of real and artificial elements in the line of sight, with the aim of highlighting important objects to improve human performance [15]. The integration of AR technology and HUD technology allows the combination of virtual and real information in the driver's primary field of view, and the reflected information appears to be part of the driving situation itself [16]. In this context, the AR-HUD system is supposed to be an effective in-vehicle display form of connected information. However, the complexity of driving behavior indicates that the design of these systems should be in accordance with the needs and preferences of drivers; otherwise, the system cannot work well and may even negatively affect driving behavior [1]. Even technologies that meet 100 % of their objectives are only effective if they are (a) accepted and (b) used [17]. Therefore, drivers' acceptance is essential for the implementation of the AR-HUD systems.

Some studies have discussed the subjective evaluation of AR-HUD systems. Schömig et al. (2018) stated that the additional AR-HUD as a supplement to the human machine interface (HMI) designed for automated driving is highly accepted, and the AR concept was rated to be more understandable and more useful [18]. Braun et al. (2018) indicated that ADAS was well accepted and that most participants perceived them as useful, reassuring, and trustworthy. Among them, usefulness was predicted by the acceptance aspects of reassurance, trust, and annoyance [19]. Smith et al. (2021) found that the methods in which AR-HUD presents the information can impact driver scanning behaviors and preferences [20]. Guo et al. (2014) used stated preference data to obtain the driver's need for driving information and their preference for information presentation and display zone images [1]. Researchers have also explored the effects that AR graphics have on a driver's distance estimation/depth perception in an augmented side view mirror, implying that AR graphics may be useful and distracting at the same time [21]. However, drivers seem inclined to trust HUDs and may prefer to use them [19,20], even when the addition of AR cues is confusing [22]. In addition, the reasons for the lack of acceptance were also discussed. Previous research analyzed the discrepancy between aging drivers' awareness and willingness to use ADAS and found that the barrier most often mentioned to prevent them from purchasing and using ADAS was a lack of perceived usefulness (32 %), followed by functional limitations, high costs (each 9 %), and trust issues (8 %) [24]. Some research reported that a possible reason for the lack of acceptance was that the focus of HUD design has been on implementing technology rather than accommodating users [1]. HUD may be considered an obstacle for driving [25], and thus increase the accident rate due to the driver's lack of visual focus [26].

Quantitative studies of drivers' acceptance of emerging technology in the connected environment such as cooperative vehicle infrastructure systems (CVIS), autonomous vehicles (AVs), and advanced in-vehicle information systems have been conducted. The studies of the above emerging technologies can provide a reference for the research on AR-HUD acceptance in connected environments. CVIS is an important component of intelligent transportation systems, which can comprehensively achieve the dynamic and real-time V2V and V2I information exchange, expand the space and time boundaries of traffic information, diversify the means for obtaining the information, and widely carry out the active safety control of the vehicle and the cooperative control of the road [27]. Zhang et al. (2021) used the technology acceptance model (TAM) to measure the public acceptance of CVIS and gender differences [28]; Li et al. (2021) established an extended TAM with four extended variables (warning service quality, distraction perception, personal

innovation, and trust) to analyze drivers' acceptance and influencing factors of CVIS [29]; AVs are those which are capable of sensing their environments in order to perform at least some aspects of the safety-critical control (like steering, throttling, or braking) without direct human input [30]. Both connected human-driven vehicles and connected automated vehicles are components of the connected environment. Xu et al. (2021) found that perceived usefulness, trust, and perceived safety were positive predictors of participants' intention to use and willingness to take AVs more than once [31]; Roberts et al. (2012) studied drivers' acceptance of two driving assistance technologies based on the TAM model with simulated driving [32]; Park et al. (2015) considered four psychological factors of car navigation systems (perceived locational accuracy, satisfaction, perceived system reliability, and service and display quality) based on foundational TAM. They showed that the service & display quality components of the systems were the most significant determinants of drivers' attitudes toward and intention to use car navigation systems [33]. Other researchers further examined the influence of drivers' age, gender, and road environment on the acceptance of ADAS through real vehicle experiments [34,35].

In general, the huge potential of AR-HUD is undeniable in the field of connected autonomous driving [36], and drivers' acceptance of and willingness to use it are key prerequisites [17,37]. It was demonstrated that drivers' characteristics, system service quality, and psychological perception in the interaction process are potential factors affecting drivers' acceptance of the emerging system. However, quantitative research on AR-HUD acceptance based on a theoretical model is still immature. It is necessary to clarify the uncertainty of drivers' acceptance of AR-HUD in the connected environment and its influencing factors. To address the above questions, this paper will comprehensively consider drivers' characteristics, system service quality, and the human-computer interaction process to examine drivers' acceptance of AR-HUD in connected environments based on TAM theory and explore the effect of potential factors. The study contributes to solving the imbalance between AR-HUD design and users' needs and realizing the potential advantages in driving safety improvement.

The rest of the article is organized as follows: Section 2 provides the details of the theoretical background and hypotheses. Section 3 describes the method. Section 4 presents the study results. Section 5 includes a discussion of the study findings, and Section 6 is the conclusion.

2. Theoretical background and hypotheses

This section aims to establish the extended TAM model to identify the determining factors of drivers' acceptance of the AR-HUD system. Based on the foundational TAM, the extended variables from drivers' characteristics, system service quality, and the human-vehicle interaction were introduced and the corresponding hypotheses are put forward to establish the extended TAM.

2.1. Foundational model

To explain users' acceptance of new technology, previous studies have developed several theoretical models, such as the Technology Acceptance Model (TAM), Theory of Planned Behavior (TPB), and Unified Theory of Acceptance and Use of Technology (UTAUT) [38]. TPB was developed to explain human behavior in general, whereas TAM and UTAUT were specifically developed to explain technology acceptance. It was demonstrated that the TAM performs best in analyzing drivers' acceptance of ADAS [37].

TAM is an adaptation of the theory of reasoned action [39] to model users' acceptance of information systems, and aims to provide a conceptualized model with a theoretical basis [40]. The foundational TAM consists of four constructs, including two determining factors to accept information technologies, namely, perceived usefulness (PU), perceived ease of use (PEOU), attitude (ATT), and intention to use (IU).

Simultaneously, additional variables can be extended based on the TAM to improve the explanatory power of the model. Davis et al. (1989) introduced definitions of these factors in the TAM [40]. They defined perceived usefulness as the level to which users believe that a specific technology or service improves overall work performance. Perceived ease of use was defined as the level at which users believe that employing a specific technology or service could be free from mental and physical effort. Attitude was referred to as the users' preference when they utilize specific technologies or systems. Intention to use was defined as the level of the user's cognitive state regarding motivation to use the particular services and technologies.

The technology acceptance model holds that system use is determined by behavioral intention, and behavioral intention is determined by attitude and perceived usefulness. The attitude is determined by the perceived usefulness and perceived ease of use. Perceived usefulness is determined by the perceived ease of use and external variables, and perceived ease of use is determined by external variables. The internal influence relationships among the factors have been confirmed [37,41]. The following hypotheses about the acceptance of AR-HUD were proposed based on foundational TAM:

- H1: Perceived ease of use has a significant positive effect on attitude;
- H2: Perceived ease of use has a significant positive effect on perceived usefulness;
- H3: Perceived usefulness has a significant positive effect on attitude;
- H4: Perceived usefulness has a significant positive effect on intention to use;
- H5: Attitude has a significant positive effect on intention to use.

2.2. Extended variables

In terms of system characteristics, the human–computer interaction process, and personal attributes, we developed five external characteristics, including trust, information service quality, display service quality, perceived distraction, and driving skills. We hypothesized that these characteristics are likely to contribute significantly to user perspectives and the adoption of AR-HUD systems. Specifically,

2.2.1. Trust (TR)

Trust was defined as the “attitude that an agent will achieve an individual's goals in a situation characterized by uncertainty and vulnerability” [42]. Here, trust is described as the degree to which the driver believes in and relies on AR-HUD in connected environments to achieve the goal of safe driving. Only drivers who trust the system are willing to use AR-HUD and follow its recommendations. Previous studies have shown that trust is a prerequisite of social behavior, especially regarding important decisions. It has been demonstrated that trust is an important factor in influencing consumers' attitudes and intentions in the context of E-commerce involved in inquiring about and purchasing books on the Internet [43,44], and it is also an important determinant for acceptance of safety-critical systems [45]. It was found that trust is the fourth most important reason for the difference in awareness and willingness of aging drivers to use ADAS [25]. Furthermore, some researchers have confirmed the influence of trust when it is considered as an indirect influence through perceived usefulness [46]. In addition, a positive effect between drivers' trust and intention to use AVs has been demonstrated [47]. Therefore, the following hypotheses are proposed:

- H6(a): Trust has a positive effect on perceived usefulness.
- H6(b): Trust has a positive effect on attitude.
- H6(c): Trust has a positive effect on intention to use.

2.2.2. Information service quality (ISQ) & display service quality (DSQ)

The definition of “service quality” was first introduced by DeLone and McLean. They defined it as the degree of the overall performance of an information service and system [48]. Considering that the HUD is a

special display device, the information service quality and interface service quality are discussed separately. This paper defines information service quality as the integrity, timeliness, and accuracy of information. Display service quality refers to the artistry and rationality of the interface layout and the effectiveness of information expression. Park and Park (2019) indicated that the functional requirements and interface design and their interplay play an essential role in utility, usability, and usefulness [49]. HUD functions determine the system information contexts and directly affect the system utility perceived by drivers. At the same time, HUD interfaces determine the usability (same definition as the PEOU in the paper) of the system. In addition, Ref. [36] mentioned that the consistency of information presentation and driver expectations directly affect the driver's trust in the system, that is, the information service quality will directly affect the trust. Another study has also shown a direct relationship between perceived service quality and the intention to use [50]. Therefore,

- H7(a): Information service quality is positively associated with perceived usefulness.
- H7(b): Information service quality is positively associated with trust.
- H7(c): Information service quality is positively associated with intention to use.
- H8(a): Display service quality is positively associated with perceived ease of use.

2.2.3. Perceived distraction (PD)

Pettitt et al. (2009) proposed a comprehensive definition for driver distraction, comprising of four components: agent, type, mechanism, and impact. The distraction was described as follows: Due to the agents of some event, activity, object, or person, within or outside the vehicle, the driver's auditory, biomechanical, cognitive, or visual faculties or combinations of these types were compromised. Following the mechanism of inducing the driver's shifting attention away from fundamental driving tasks, the driver was impacted and is delayed in the recognition of information necessary to safely maintain the lateral and longitudinal control of the vehicle or the driving task [51]. A HUD could direct gaze toward the road, but overlaying the windshield can also degrade driving performance [52], cause a loss of global situation awareness [53], and cause inattentive blindness [54]. Some studies reported potential safety concerns associated with the use of HUDs, such as visual clutter and cognitive capture [55–57]. With more functionality, more and more information will be generated and make the driving environment very complicated, which makes it very difficult for drivers to digest and respond to all of the information [58]. Displaying too much information through HUDs can result in information overload. Therefore, perceived distraction is defined as the interference of information interaction on the driver's visual, auditory, and perceptual behaviors in the process of driving. When such interference is considered to negatively impact the main driving task and driving safety, the usefulness, ease of use, and trust in the system will decrease. Therefore, this paper assumes that perceived usefulness has a negative effect on trust, perceived usefulness, and perceived ease of use.

- H9(a): Perceived distraction is negatively associated with perceived usefulness.
- H9(b): Perceived distraction is negatively associated with perceived ease of use.
- H9(c): Perceived distraction is negatively associated with trust.

2.2.4. Driving skills (DS)

The term “driving skills” is defined as drivers' subjective evaluation of their ability to drive safely (e.g., emergency response and fluent car control). How drivers perceive their driving skills affects their safety strategies [59], such as adjusting driving behavior and adopting a driving assistance system. Additionally, participants reported their preference through perceived usability and acceptance, and confirmed

that the dissociation of performance and preference depends on participants' skills [60]. Individuals have disparate information needs according to their driving skills, and poorer driving abilities may be associated with more positive perceptions of the system since individuals with compromised driving skills have more to gain from the technology [17]. With driving, it may be the case that through experience, fewer conscious resources are required for controlling the vehicle as driving skill becomes automatic and this frees up resources [61]. That means drivers with good driving skills usually have more remaining resources to complete other tasks in driving. Meanwhile, high levels of driving skills are conducive to the driver's ability to deal with the information input around the vehicle integrally [62]. Therefore, it's inferred those drivers with good driving skills usually have more remaining resources and better information processing capability to interact with AR-HUD, influencing the perceived ease of use of AR-HUD. Thus, we inferred that driving skills would affect drivers' evaluations of the perceived usefulness and perceived ease of use of the system.

H10(a): Driving skills are associated with perceived usefulness.

H10(b): Driving skills are associated with perceived ease of use.

Finally, the proposed conceptual model of this study is depicted in Fig. 1.

3. Method

3.1. Questionnaire design

A self-administered questionnaire was designed to collect empirical data. The questionnaire consists of three sections. The first section measures respondents' information with seven items, namely, gender, age, driving experience, driving frequency, educational background, familiarity with HUD, and user experience.

The second section introduced the AR-HUD in a connected environment with a 90-second video presenting the design concept, information types, and typical application scenarios. Referring to the classification of commercial HUD information and the definition of different driving tasks in previous studies [49,63], the information types were grouped into four categories: vehicle state, safety, navigation, and communication/infotainment/environment. The vehicle state category consists of speed, cruise control-related information, gear shift-related information, RPM, system messages, fuel-related information, turn signal status, etc. The safety category is composed of a collision warning, road sign notification, lane keeping-related information, night vision-related warning, etc. The navigation category comprises navigation instructions, the remaining distance to destination, compass heading, etc. The fourth information category includes radio, audio player, phone call, and outside temperature. The typical application scenarios included speed, AR navigation, lane enhancement, forward collision

warning, phone call reminders, etc. Several examples of application scenarios are shown in Fig. 2.

The third section measured respondents' perceptions about usefulness, ease of use, attitude, intentions to use, trust, display service quality, information service quality, and driving skills. All items were measured with a five-point Likert-type scale, ranging from "strongly disagree (=1)" to "strongly agree (=5)". The questionnaire was carefully designed and was controlled at 3–4 items in each variable measurement to avoid the abnormally high value of Cronbach's α in reliability analysis caused by a large number of the items. Combined with the system characteristics, different dimensions were also designed to avoid similar questions in variable measurement. Next, the experts' argument and small sample test were conducted, and the initial questionnaire was revised based on the feedback to avoid ambiguity or redundancy in the items. The items in various constructs in the final questionnaire are shown in Table 1.

3.2. Participants and data collection

Since the number of consumers of AR-HUD products is unknown, a convenience sampling technique was used to collect online questionnaires. In this questionnaire, participants provided their demographic information and answered 29 items in 9 structures. We excluded responses that were obviously inaccurate based on these five aspects to guarantee the quality and validity of the data: (1) Participants must have a driving license and driving experience over the past year. (2) The duration of the video must be longer than 95 s. (3) The answering time on the specific page must be longer than 30 s. (4) The overall answering time must not be <300 s. (5) The answers on consecutive pages are not completely consistent. Participants submitting valid questionnaires are paid 6–8 yuan per person.

Finally, 388 valid questionnaires were collected. The drivers' information is shown in Table 2. The distribution of participants' gender, age, driving experience, driving frequency, and educational background were shown. In addition, the participants' familiarity with HUD and user experience were also investigated.

3.3. Data analysis

Data analysis was completed by Amos and SPSS. In the first stage, we assessed the reliability and validity of the measurement model based on internal consistency, convergent validity, and discriminant validity. In the second stage, structural equation modeling (SEM) was used to verify the hypothetical model framework to explore the relationship between variables and the driver's acceptance of the AR-HUD.

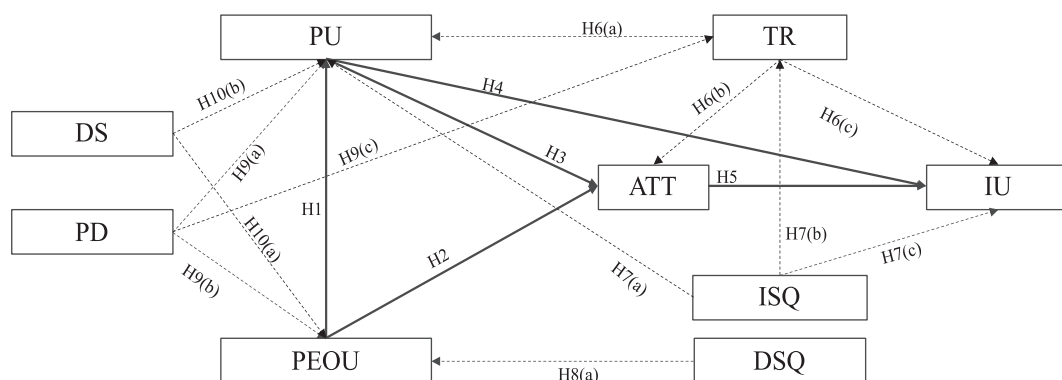
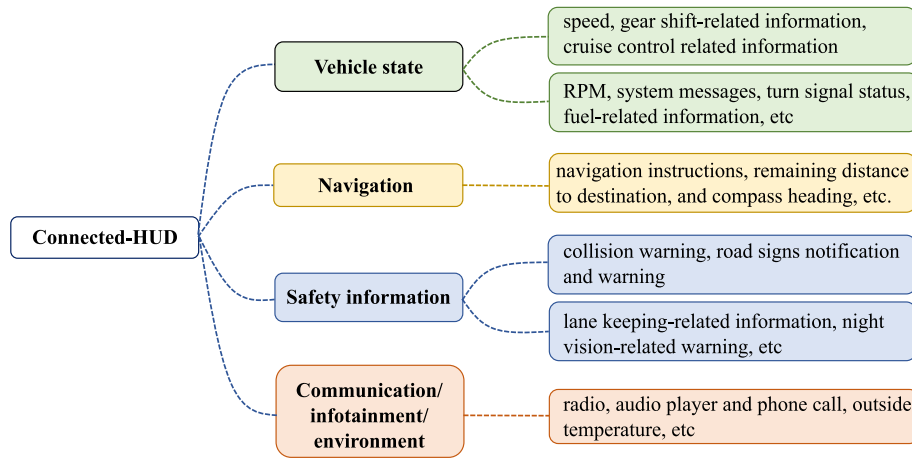


Fig. 1. Extended TAM.



(a) Information types



Fig. 2. Examples of application scenarios in video.

4. Results

4.1. Measurement model

The measurement model should be assessed before the structural model is examined, including reliability analysis and validity analysis. Reliability refers to the internal consistency and reliability of measurement results. Cronbach's α is used to validate internal consistency, and 0.7 or higher is recommended. All Cronbach's α values exceed 0.7, showing good reliability.

Validity refers to the degree to which the measurement model can accurately measure the target, including structural validity, convergent validity, and discriminant validity. To assess the structural validity, confirmatory factor analysis was performed using AMOS 24.0. χ^2 test and its fitting index (e.g., Root Mean Square Error of Approximation (RMSEA), Goodness-of-Fit Index (GFI), Normed Fit Index, (NFI), Comparative Fit Index (CFI), and Incremental Fit Index (IFI)) were used to examine the fitting effect. The χ^2/df should be in the range of [1–3], the fitting index should be greater than or equal to 0.90, and RMSEA should be <0.08 . The results show that $\chi^2/df = 1.736$, GFI = 0.908, RMSEA = 0.044, NFI = 0.947, CFI = 0.976, IFI = 0.976, and all the values meet the evaluation criteria.

Convergent validity was assessed based on item loadings, composite reliability (CR), and the average variance extracted (AVE). Item loadings are recommended to exceed 0.6 [68]. The composite reliability is recommended to exceed 0.7. The average variance extracted (AVE) value for each latent variable should exceed 0.5, and the square root of the AVE should be greater than the inter-construct correlations [69]. As shown in Table 3, all item loadings, all AVEs, and the composite reliability exceeded 0.6, 0.5, and 0.7, respectively. These results showed good convergent validity [70].

To assess discriminant validity, we compared the square root of AVE for each factor to its correlations with other factors. As we can see from Table 4, the square root of AVE for each factor is obviously larger than its correlation coefficients with other factors. Thus, the scale has good

discriminant validity [70]. In addition, the correlation values between latent variables were lower than 0.5, so multicollinearity issues were avoided. Thus, the measurement model is proven to be reliable and valid for the study.

4.2. Structural model

The structural model comprises the hypothesized relationship between latent constructs in the research model. The value of the fitting index is $\chi^2/df = 1.838$, GFI = 0.903, RMSEA = 0.047, NFI = 0.943, CFI = 0.973, and IFI = 0.973. Overall, the predictive validity and model fit indices were satisfactory.

Table 5 shows parameter estimates and the hypothesis results of the extended TAM model. The relationship between variables is expressed by path coefficients. The value of the path coefficient can effectively represent the influence degree between different variables. A positive value represents a positive correlation, while a negative value represents a negative correlation. Finally, the standardized path coefficients and significant results between each latent variable are obtained. The results showed that all hypotheses (H1, H2, H3, H4, H5) introduced in the foundational TAM about variables (PEOU, PU, ATT, and IU) are supported. For extended variables (TR, ISQ, DSQ, PD, and DS), all hypotheses except for H7 (a) and H9 (a) are supported. The results are shown in Table 5 and Fig. 3.

5. Discussion

This study examined the factors affecting drivers' use intentions of AR-HUD products based on the extended TAM. On the one hand, this paper explores the impact of original variables and extended variables on drivers' acceptance on the basis of verifying the effectiveness and adaptability of the TAM. On the other hand, some suggestions are put forward for the design and promotion of HUD. In addition, the limitations of the study and areas of future research are pointed out.

Table 1
Construct and items.

Construct and item	Source
Attitude ATT1: Using AR-HUD in my trip is a good idea. ATT2: I like to use AR-HUD in driving. ATT3: Using AR-HUD in my trip is beneficial to me.	Adapted from [64,65]
Perceived usefulness PU1:AR-HUD can help me obtain the surrounding traffic situation. PU2:AR-HUD can help me operate the vehicle more efficiently. PU3:AR-HUD can help me identify the hazards in the process of driving.	Adapted from [40,66]
Perceived ease of usefulness PEOU1:AR-HUD is very simple and convenient to use. PEOU2: I can quickly master the usage of AR-HUD. PEOU3: I think the amount of information provided by AR-HUD is easy to accept.	Adapted from [40]
Intention to Use IU1: I will buy the AR-HUD system if I can afford it. IU2: I want to drive a vehicle with AR-HUD system. IU3: I will use AR-HUD system when I am driving a vehicle equipped with the system. IU4: I would recommend my friend to use AR-HUD while driving.	Adapted from [64,67]
Trust TR1: I feel safer when using the AR-HUD. TR2: I will make corresponding driving operations according to the tips of AR-HUD. TR3: I can achieve safe driving relying on AR-HUD	Adapted from [66,46,31]
Information Service Quality ISQ1: I think the information provided by the system is complete. ISQ2: I think the information provided by the system can meet my demands. ISQ3: I think the information provided by the system is timely.	Adapted from [50]
Display Service Quality DSQ1: I think the design of the system interface is beautiful (attractive). DSQ2: I think the location and size of the system information are reasonable. DSQ3: I think the display mode of the system information is appropriate.	Adapted from [50]
Perceived Distraction PD1: The interface will interfere with my observation of the road ahead. PD2: The voice prompts in the system will interfere with the obtain of other auditory information. PD3: The system's prompts will distract my attention. PD4: The system's prompts will interfere with my judgment.	Self-developed
Driving Skill DS1: I can predict the potential danger of driving. DS2: I can make quick response and decision in case of emergencies. DS3: I can operate the vehicle efficiently and drive smoothly.	Adapted from [59]

Table 2
Demographic characteristics of the participants (N = 388).

Category	Levels	Observations	Percentage
Gender	male	225	57.99 %
	female	163	42.01 %
Age	<=20	6	1.55 %
	<=30	113	29.12 %
	<=40	113	29.12 %
	<=50	92	23.71 %
	>50	64	16.49 %
Driving experience	1–3 years	82	21.13 %
	3–5 years	57	14.69 %
	5–10 years	102	26.29 %
	10–20 years	104	26.80 %
	>20 years	43	11.08 %
Driving frequency	almost every day	175	45.10 %
	3–5 times per week	70	18.04 %
	1–2 times per week	60	15.46 %
	less than once per week	83	21.40 %
Educational background	master degree or above	72	18.56 %
	junior college / undergraduate	192	49.48 %
	high school	82	21.13 %
	junior middle school or below	42	10.83 %
Familiarity with HUD	have never heard	182	46.91 %
	have some knowledge	167	43.04 %
	have a good understanding	39	10.05 %
User experience	Yes	15	3.87 %
	No	373	96.13 %

the foundational TAM are supported [40]. Specifically, perceived ease of use significantly affects perceived usefulness ($H1, \beta = 0.736, p < 0.001$), perceived ease of use and perceived usefulness jointly affect attitude ($H2, \beta = 0.553, p < 0.001$; $H3, \beta = 0.401, p < 0.001$), and attitude and perceived usefulness jointly affect usage intention ($H4, \beta = 0.402, p < 0.001$; $H5, \beta = 0.147, p = 0.002$). It can be concluded that perceived usefulness and perceived ease of use are still the key factors determining the acceptance of the AR-HUD system. Perceived ease of use significantly affects perceived usefulness and attitude. The result is similar to another in-vehicle information technology study, which demonstrated that participants' perceived ease of use of in-vehicle global positioning system (GPS) products has a significant, positive effect on perceived usefulness and attitude [65]. In contrast, perceived ease of use was a weak predictor of the acceptance of AVs [71]. One possible reason for the mixed results could be that the role of perceived ease of use was related to the different system types and the different interaction progress between participants and the system [71]. With respect to the effort demand for the system interaction, little effort was required when riding in autonomous vehicles; however, drivers need to continuously interact with AR-HUD while driving to process and respond to the driving-related information. Therefore, the continuous effort demanded gives perceived ease of use a more significant impact on the acceptance of the AR-HUD system. At the same time, perceived usefulness has the greatest impact on usage intention, while other studies also reported the dominant role of perceived usefulness in shaping drivers' intentions to use in-vehicle information and automation technologies [37,50]. Next, by observing the relationships of the extended variables and the original variables, the key factors that determine the driver's acceptance of the AR-HUD system can be further identified. Further analysis of the extended variables is as follows.

5.1. Influence analysis of variables

The results show that the extended TAM model has good adaptability to measure drivers' acceptance of AR-HUD. All hypotheses introduced in

Table 3
Reliability and validity tests of extended TAM questionnaire.

Construct	Item	Item loading	Cronbach's α	AVE	CR
Perceived ease of usefulness	PEOU1	0.87	0.89	0.69	0.87
	PEOU2	0.81			
	PEOU3	0.81			
Perceived usefulness	PU1	0.89	0.92	0.79	0.92
	PU2	0.91			
	PU3	0.87			
Attitude	ATT1	0.93	0.94	0.83	0.94
	ATT2	0.93			
	ATT3	0.88			
Intention to use	IU1	0.87	0.94	0.80	0.94
	IU2	0.91			
	IU3	0.90			
	IU4	0.90			
Information service quality	ISQ1	0.76	0.89	0.69	0.87
	ISQ2	0.89			
	ISQ3	0.83			
Display service quality	DSQ1	0.87	0.92	0.80	0.92
	DSQ2	0.91			
	DSQ4	0.90			
Perceived distraction	PD1	0.81	0.94	0.79	0.94
	PD2	0.89			
	PD3	0.93			
	PD4	0.91			
Trust	TR1	0.91	0.87	0.71	0.88
	TR2	0.81			
	TR3	0.80			
Driving skill	DS1	0.79	0.89	0.72	0.89
	DS2	0.93			
	DS3	0.83			

Trust (TR). The importance of trust has been confirmed in different domains, especially in the adoption of new technologies [70]. It is suggested that trust may explain the individual acceptance of driving assistance systems [72]. In addition, trust is a major determinant of reliance on and acceptance of automation technology, standing between people's beliefs toward automation and their intention to use it [73,74]. In line with previous research, it was demonstrated that trust exhibited strong direct effects on intention to use ($H6(c)$, $\beta = 0.179$, $p = 0.019$) and indirect impacts on intention to use through perceived usefulness ($H6(a)$, $\beta = 0.246$, $p < 0.001$) and attitude ($H6(b)$, $\beta = 0.143$, $p = 0.03$).

Table 4
Extended TAM related variables statistics.

	Mean	SD	PEOU	PU	ATT	IU	ISQ	DSQ	PD	TR	DS
PEOU	4.16	0.76	(0.83)								
PU	4.26	0.77	0.39***	(0.89)							
ATT	4.07	0.88	0.40***	0.46***	(0.91)						
IU	4.16	0.83	0.42***	0.45***	0.48***	(0.90)					
ISQ	4.06	0.74	0.35***	0.37***	0.38***	0.45***	(0.83)				
DSQ	4.06	0.75	0.36***	0.35***	0.33***	0.39***	0.42***	(0.90)			
PD	3.15	1.07	-0.28***	-0.29***	-0.38***	-0.38***	-0.32***	-0.28***	(0.89)		
TR	3.85	0.78	0.36***	0.39***	0.40***	0.47***	0.45***	0.41***	-0.42***	(0.84)	
DS	3.74	0.88	0.16***	0.107**	0.13***	0.15***	0.17***	0.16***	-0.05***	0.17***	(0.85)

Note:

*** < 0.01 ,

** < 0.05 .

Moreover, it mediates the relationship between information service quality and perceived usefulness. The results indicate that only when the information is trusted can the driver feel that it is useful for driving. All the findings proved that trust is a key factor and precondition of drivers' acceptance and intention to use connected AR-HUD.

Calibrating the trust of highly diverse users to an appropriate level will help distrusting drivers to more comfortably experience the benefits of AR-HUD [75]. Previous studies demonstrated that augmented reality assistance has the potential to increase user acceptance and trust by communicating system decisions, (i.e., transparent system behavior) to the user as the system proves its awareness of its surroundings [45,76,77]. In addition, a comprehensive level of understanding of the technology seemed to have a positive effect on both acceptability and trust of the technology [78].

Information service quality & Display service quality (ISQ&DSQ).

The usability is mainly determined by the HUD interfaces resulting from screen-level interface design. Functional requirements analysis, that is, the specification of HUD functions directly affects the system utility perceived by the drivers [49]. The results show that display service quality significantly affects perceived ease of use ($H8(a)$, $\beta = 0.315$, $p = 0.001$), thus affecting the intention to use, which is consistent with the above research [49]. However, a direct effect between information service quality and perceived usefulness was not observed ($H7(a)$, $\beta = 0.029$, $p = 0.767$). Information service quality indirectly affects perceived usefulness through its strong effect on trust ($H7(b)$, $\beta = 0.953$, $p < 0.001$), and has a direct impact on intention to use ($H7(c)$, $\beta = 0.315$, $p = 0.001$). The results help to further understand the roles of functional requirements analysis and interface design mentioned above and their interplay in the design of automotive HUDs. It can be inferred that although HUD boasts rich functions, only when the information is trusted can it be deemed useful by drivers. Then, in addition to expanding the scope of functions, timeliness, effectiveness, accuracy, and consistency with the driver's expectations of information (i.e., information service quality) are important factors affecting HUD acceptance. Therefore, on the basis of clarifying drivers' information needs, the service quality of HUD can be comprehensively improved by providing it to drivers at the right time through reasonable interface design.

Perceived distraction (PD). The results show that perceived distraction has a significant negative effect on perceived ease of use ($H9(b)$, $\beta = -0.099$, $p < 0.001$) and trust ($H9(c)$, $\beta = -0.16$, $p < 0.001$) and has no significant effect on perceived usefulness ($H9(a)$, $\beta = 0.035$, $p = 0.204$). Exactly as the negative effect of visual enhancement design on driver's tracking errors and subjective workloads was found in previous studies [56,57], participants reported that the AR-HUD caused a certain amount of distraction. However, the negative evaluation did not influence participants' perception of the usefulness but mainly lies in the ease of use and trust of the system and has a greater negative impact on trust. Previous studies provide evidence that individual preferences such as willingness to readily adopt new technology might predispose drivers to

Table 5
Extended TAM related variables statistic.

Hypothesis	Path	Path coefficient	S.E.	C.R.	P	Result
H1	PEOU- →PU	0.736	0.074	9.919	<0.001	Supported
H2	PEOU- →ATT	0.553	0.121	4.555	<0.001	Supported
H3	PU- →ATT	0.401	0.119	3.357	<0.001	Supported
H4	PU- →IU	0.402	0.073	5.514	<0.001	Supported
H5	ATT- →IU	0.147	0.047	3.144	0.002	Supported
H6(a)	TR- →PU	0.246	0.073	3.353	<0.001	Supported
H6(b)	TR- →ATT	0.143	0.066	2.175	0.03	Supported
H6(c)	TR- →IU	0.179	0.077	2.337	0.019	Supported
H7(a)	ISQ- →PU	0.029	0.099	0.296	0.767	Rejected
H7(b)	ISQ- →TR	0.953	0.068	14.08	<0.001	Supported
H7(c)	ISQ- →IU	0.315	0.097	3.243	0.001	Supported
H8(a)	DSQ- →PEOU	0.715	0.052	13.805	<0.001	Supported
H9(a)	PD- →PU	0.035	0.027	1.269	0.204	Rejected
H9(b)	PD- →PEOU	-0.099	0.027	-3.621	<0.001	Supported
H9(c)	PD- →TR	-0.16	0.03	-5.382	<0.001	Supported
H10(a)	DS- →PU	-0.081	0.027	-3.001	0.003	Supported
H10(b)	DS- →PEOU	0.078	0.031	2.506	0.012	Supported

Note: S.E.: Standard error; C.R.: Critical ratio.

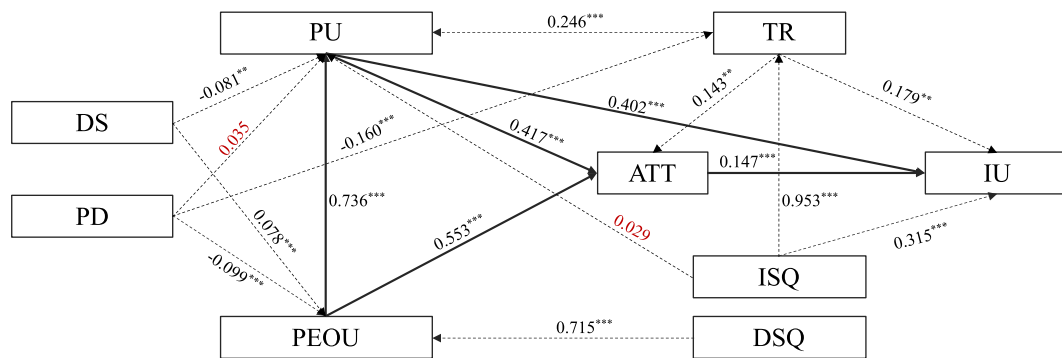


Fig. 3. Path Result of Extend TAM Model.

use HUDs even though it may diminish their driving abilities [22,23,79]. Therefore, reducing visual and perceptual distraction through interface design and information flow rate will help to improve drivers' trust in AR-HUD.

Driving skills (DS). Poor driving skills are related to aberrant driving behaviors (i.e., violations and errors), which are dangerous and may cause an accident [59]. The results demonstrated that driving skills have a positive effect on perceived ease of use (10(b), $\beta = 0.078$, $p = 0.012$) and negatively affect perceived usefulness (H10(a), $\beta = -0.081$, $p = 0.003$). One possible explanation for the negative effect is that the technology may challenge the extensive driving experience that they have acquired over the years, even though drivers need to develop new behavioral patterns [80]. Another possible explanation is that some drivers view ADAS as a threat to their self-image as a good driver rather than a benefit that could make their life easier. This attitude might be reflected in an unrealistically optimistic assessment of their driving skills and a lower intention to use ADAS [1]. This finding indicated that drivers with poor driving skills, such as new drivers and aging drivers, are potential users of AR-HUD systems.

5.2. Suggestions for AR-HUD designer

The lack of public awareness and acceptance of AR-HUD hinders the popularization and advantages of the system [78]. Based on the potential role of various variables affecting drivers' acceptance of AR-HUD, this paper provides suggestions to improve drivers' perceived usefulness and perceived ease of use of AR-HUD.

In terms of perceived usefulness, the results show that trust directly affects perceived usefulness, and information service quality also affects

perceived usefulness through trust. At the same time, both trust and information service quality have a direct impact on intention to use. Information service quality and trust are the key factors to improve drivers' perceived usefulness and acceptance. First, in the aspect of information service, studies have indicated that drivers want personalization and have diverging content preferences [81]. The integration information should increase sensitivity to the dynamic, context-sensitive, and individual-specific nature of the driver information needs [80]. Second, the integration of different functions in the vehicle should work together to avoid negative safety consequences, such as fighting for attention or giving conflicting information [80]. AR-HUD can be utilized to display novel or unconventional information to enhance the driving experience rather than being positioned as an additional complement for conventional displays. Third, to reduce perceived distractions and improve trust, the information design must achieve a reasonable information flow rate without exceeding human cognition capacities and reduce interference with the driver's visual, auditory, and perceptual behavior [82]. Fourth, previous research indicated that a comprehensive level of understanding of the system seemed to have a positive effect on both acceptance and trust in the technology [78]. Thus, authorities can publicize the safety improvement performance of AR-HUD to enhance the public's trust.

In terms of the perceived ease of use, the results indicate that it is affected by display service quality and perceived distraction. Therefore, improving the display service quality and reducing the perceived distraction could be effective ways to enhance drivers' perceived ease of use. As measured in the questionnaire, we should pay attention to the artistry of the interface, interface layout, and information presentation form in the aspect of interface design. (a) For the artistry of the interface,

some participants commented that the shape dimension and shape of visual cues expect a combination of design aesthetic and careful placement of visual cues to result in improved user preference [22]. (b) For the interface layout, good interface design would to some extent enable the support of a variety of information types without creating safety problems. Skrypchuk et al. (2019) reconceptualized situation awareness (SA), and the framework facilitates the creation of in-vehicle information systems that help drivers succeed in multigoal multitasking situations [83]. Merenda et al. (2018) demonstrated that vehicle- and world-animated approaches using directional and motion-based cues can support accurate judgments with a less spatial error of localization than either traditional screen-fixed or world-fixed conformal cues anchored in place [84]. Researchers have proposed the provision of “attentive user interfaces”, which are “UIs that are sensitive to the user’s attention” and “structure communication such that the limited resource of attention is allocated optimally across the user’s tasks” [85]. Additionally, both context switching and focal distance switching are important AR user interface design issues, which should be carefully designed to avoid visual fatigue and driving performance reduction [86]. (c) For the information presentation form, it is suggested that the representation of the information should be designed with a good self-explanatory ability and be easy to understand.

Furthermore, considering the effect of drivers’ skills on perceived usefulness and perceived ease of use, novice, and aging drivers seemed to be potential consumers of AR-HUD, as individuals with compromised driving skills have more to gain from the technology [17].

5.3. Limitations and future research

The present study of AR-HUD was constrained by several limitations. First, to achieve a balance between time consumption and information transmission, the video used to introduce the AR-HUD only shows typical application scenarios. However, there are diverse AR-HUD products with various functions and practical application scenarios. Thus, the AR-HUD in this paper is a representative case used to identify the key factors affecting acceptance, but the ratings of the variables may not be applicable to all types of AR-HUD. Second, as AR-HUD is not widely available, it is likely that only a small portion of the participants have prior experience with the technology. Third, although previous research indicated that participants could imagine the technology based on the provided description and pictures [87], another study revealed that participants exposed to the driving simulator showed a significantly higher intention to use these systems compared to participants in the online survey [37]. Ref. [42] indicated that perceived ease of use becomes a significant predictor of intention to use after the participants directly experienced the AVs. In addition, the subjective questionnaire was used to measure the driver’s driving skills, which will deviate from the driver’s real performance. Therefore, field studies or simulator experiments should be considered to further study the influence of user experience on the acceptance of AR-HUD and practical driving skills. Simultaneously, it is worthwhile to consider the influences of individual-related factors (e.g., age, gender, education background, income) on drivers’ perception and acceptance of AR-HUD systems. Furthermore, research needs to comprehensively consider the personalized and situational information needs, interface preferences, and driving safety impacts in connected environments to form AR-HUD design guidance.

6. Conclusions

To address the uncertainty of drivers’ acceptance of AR-HUD and its influencing factors, this study investigated drivers’ acceptance of AR-HUD by the extended TAM with additional variables (information service quality, display service quality, perceived distraction, trust, and driving skills). The reliability and validity of the measurement model are verified, and relationships between the observed variables are measured by the structural equation model.

The results indicate that (1) drivers’ trust has a positive effect on perceived usefulness, attitude, and intention to use; (2) display service quality positively affects perceived ease of use, while information service quality has a positive effect on trust and intention to use; and (3) perceived distraction has a significantly negative effect on perceived ease of use and trust, and the negative effect on perceived usefulness was not significant. Finally, (4) drivers’ subjective driving skills have a positive effect on perceived ease of use and a negative effect on perceived usefulness.

The further analysis attempts to explain the causes of the influence relationship between variables and provides suggestions to enhance the drivers’ perceived usefulness and perceived ease of use of AR-HUD. The information design is required to consider the sensitivity of information needs, information integration mode, and reasonable information flow. The interface design should comprehensively consider the artistry of the interface, interface layout, and information presentation form. At the same time, enhancing the public’s awareness and trust in the AR-HUD system is necessary, especially for potential users such as novices and aging drivers.

The current study offered insight into the uncertainty and influencing factors of drivers’ acceptance of AR-HUD in theory and supported the development of AR-HUD systems, especially in emerging markets. The findings contributed to improving drivers’ acceptance of AR-HUD and giving full play to AR-HUD’s advantages in connected and autonomous vehicles.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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